

Journey to the correct result

1st March-9th May, 2016

Subhadip Das
HWRE LAB IITK

Experimental procedure

Primary objective:

To study the location of maximum scouring under pressure flow condition.

Specifications:

Details of Flume	Length=15m, width=30cm, height=45cm, bed slope=0.
Length of test section	10m
Material used in test section	Sand
Sand characteristics	Grade-III, Ahmedabad sand
Initial sand bed depth	12cm
Bridge material	Plexi glass sheet (10mm thick)
Details of bridge	Length=15.3cm, Width=30cm
Non-dimensional parameters	$\frac{q}{Y\sqrt{gY}}, \frac{H_b}{Y}, \frac{V_a}{V_c}$

Sieve analysis:

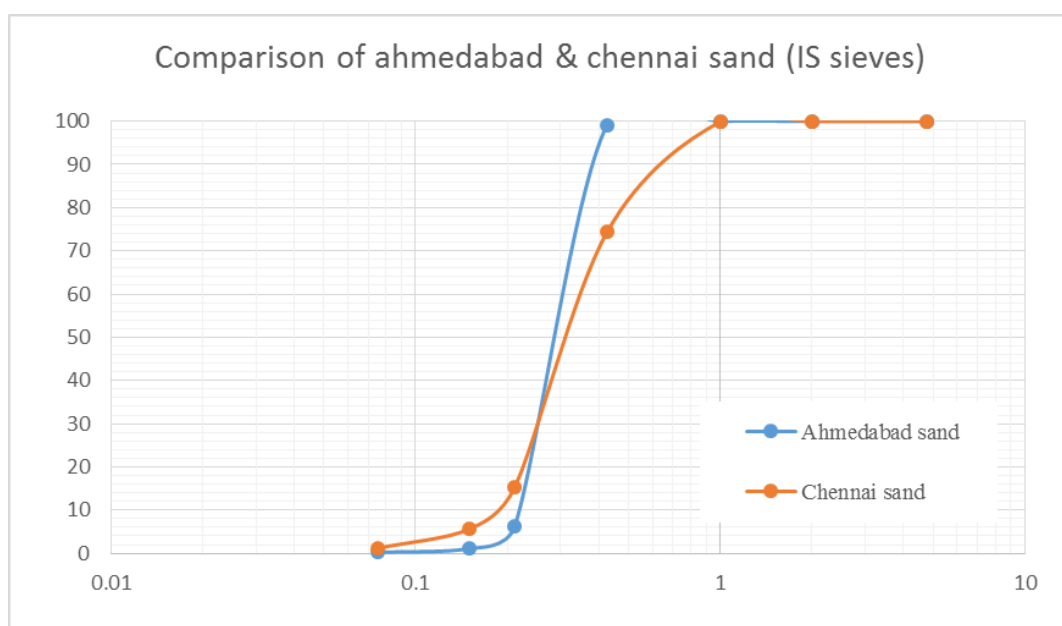
Sieve analysis is done using IS sieves & ASTM sieves for Ahmedabad & Chennai sand to visualize their grain size distribution.

IS sieves

Sieve analysis of ahmedabad sand (IS sieves)			
IS sieve No. (mm)	Weight retained (gms)	Cumulative Weight retained (gms)	% weight passing (%)
4.75	0	0	100
2	0	0	100
1	0	0	100
0.425	4.5	4.5	99.09
0.212	463.2	467.7	6.40
0.15	26.3	494	1.14
0.075	4.6	498.6	0.22
Pan	1.1	499.7	0
Total	499.7	-	-

Sieve analysis of chennai sand (IS sieves)			
IS sieve No. (mm)	Weight retained (gms)	Cumulative Weight retained (gms)	% weight passing (%)
4.75	0	0	100
2	0	0	100
1	0.6	0.6	99.88
0.425	127.6	128.2	74.32
0.212	294.1	422.3	15.41
0.15	48.9	471.2	5.61
0.075	22	493.2	1.20
Pan	6	499.2	0
Total	499.2	-	-

Comparison of ahmedabad & chennai sand (IS sieves)		
IS sieve No. (mm)	% weight passing (%)	
	Ahmedabad sand	Chennai sand
4.75	100	100
2	100	100
1	100	99.88
0.425	99.1	74.32
0.212	6.40	15.40
0.15	1.14	5.61
0.075	0.22	1.20
Pan	0	0
Total	-	-

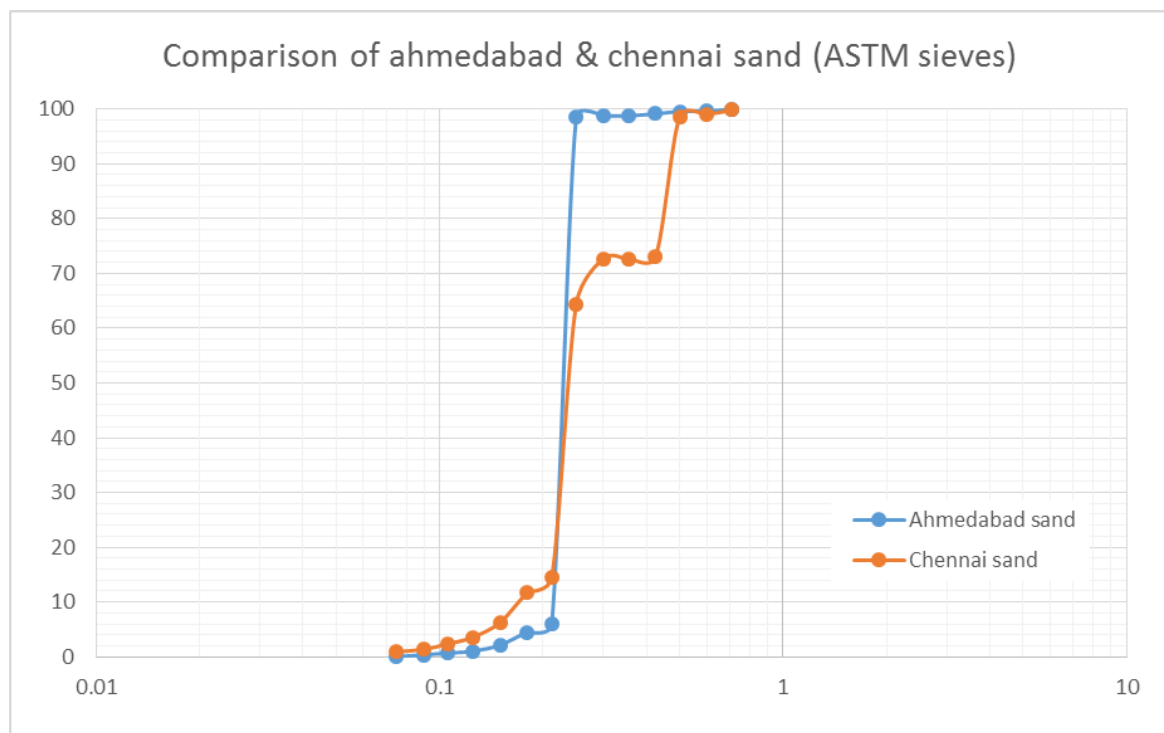


ASTM sieves

Sieve analysis of ahmedabad sand (ASTM sieves)			
IS sieve No. (mm)	Weight retained (gms)	Cumulative Weight retained (gms)	% weight passing (%)
0.71	0.03	0.03	99.99
0.6	1.9	1.93	99.61
0.5	0.33	2.26	99.55
0.425	2.06	4.32	99.13
0.355	1.8	6.12	98.77
0.3	0.18	6.3	98.74
0.25	0.67	6.97	98.60
0.212	462.77	469.74	5.94
0.18	7.07	476.81	4.53
0.15	11.82	488.63	2.16
0.125	5.52	494.15	1.05
0.106	1.67	495.82	0.72
0.09	1.73	497.55	0.37
0.075	1.44	498.99	0.09
Pan	0.43	499.42	0
Total	499.42		

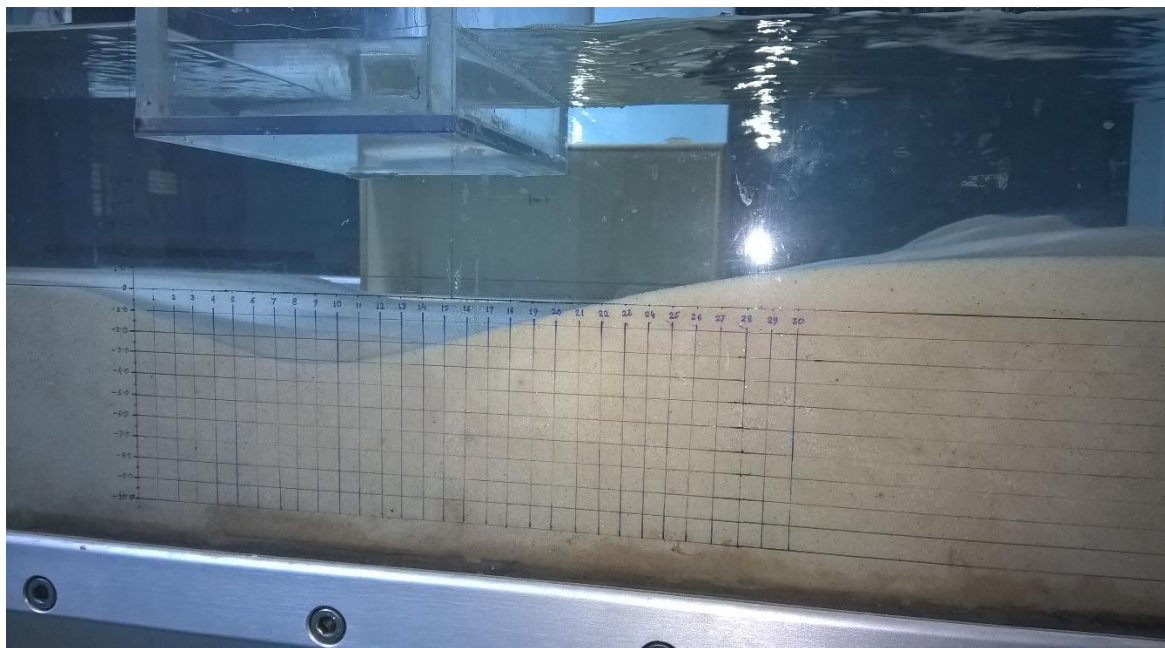
Sieve analysis of chennai sand (ASTM sieves)			
IS sieve No. (mm)	Weight retained (gms)	Cumulative Weight retained (gms)	% weight passing (%)
0.71	0.84	0.84	99.83
0.6	4	4.84	99.02
0.5	2.03	6.87	98.61
0.425	127.09	133.96	72.98
0.355	1.36	135.32	72.70
0.3	0.01	135.33	72.70
0.25	41.16	176.49	64.40
0.212	247.57	424.06	14.46
0.18	13.78	437.84	11.68
0.15	26.72	464.56	6.29
0.125	13.34	477.9	3.60
0.106	5.96	483.86	2.39
0.09	5.02	488.88	1.38
0.075	2.2	491.08	0.94
Pan	4.65	495.73	0
Total	495.73		

Comparison of ahmedabad & chennai sand (ASTM sieves)		
IS sieve No. (mm)	% weight passing (%)	
	Ahmedabad sand	Chennai sand
0.71	99.99	99.83
0.6	99.61	99.02
0.5	99.55	98.61
0.425	99.13	72.98
0.355	98.77	72.70
0.3	98.74	72.70
0.25	98.60	64.40
0.212	5.94	14.46
0.18	4.53	11.68
0.15	2.16	6.29
0.125	1.05	3.60
0.106	0.72	2.39
0.09	0.37	1.38
0.075	0.09	0.94
pan	0	0
Total		

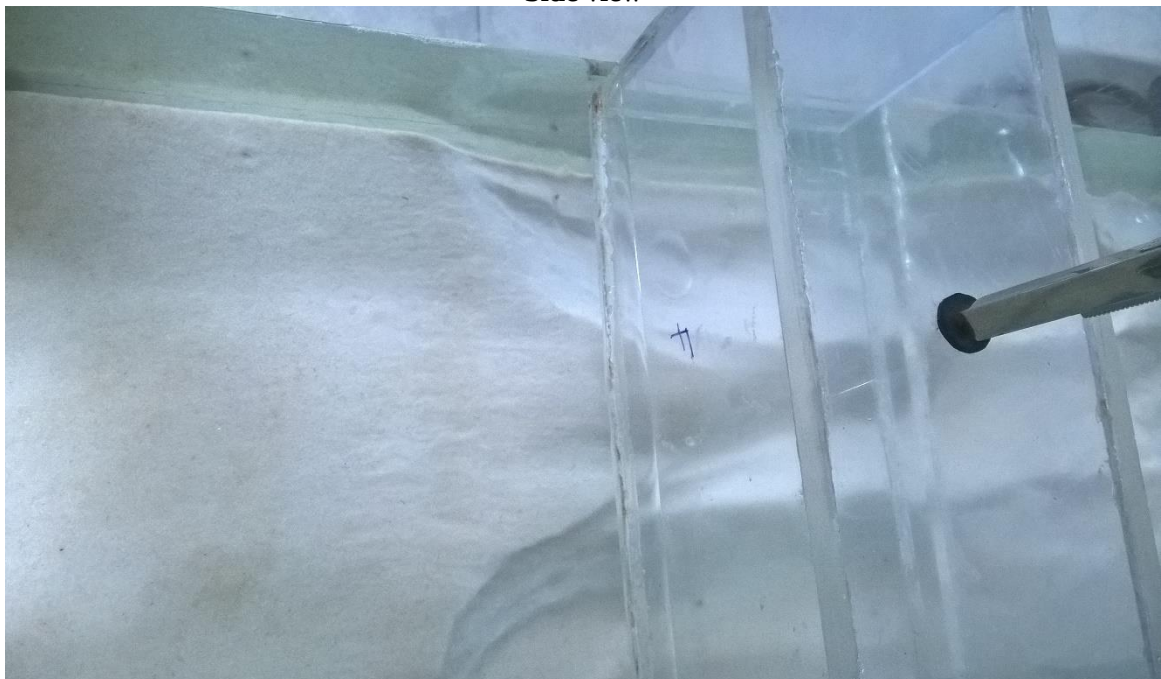


Experiment-01_(a),(b),(c)

Parameters	Experiment-01_(a),(b),(c)	Shivendra's experiment
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.212, \frac{H_b}{Y} = 0.6, \frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.212, \frac{H_b}{Y} = 0.6, \frac{V_a}{V_c} = 0.84$
Water depth	12 cm	12 cm
Approach velocity	23 cm/s	23 cm/s
Discharge	8.28 l/s	10.6 l/s
Bridge opening	7.2 cm	7.2 cm
Max. scour location	Under the bridge	d/s of the bridge



Side view



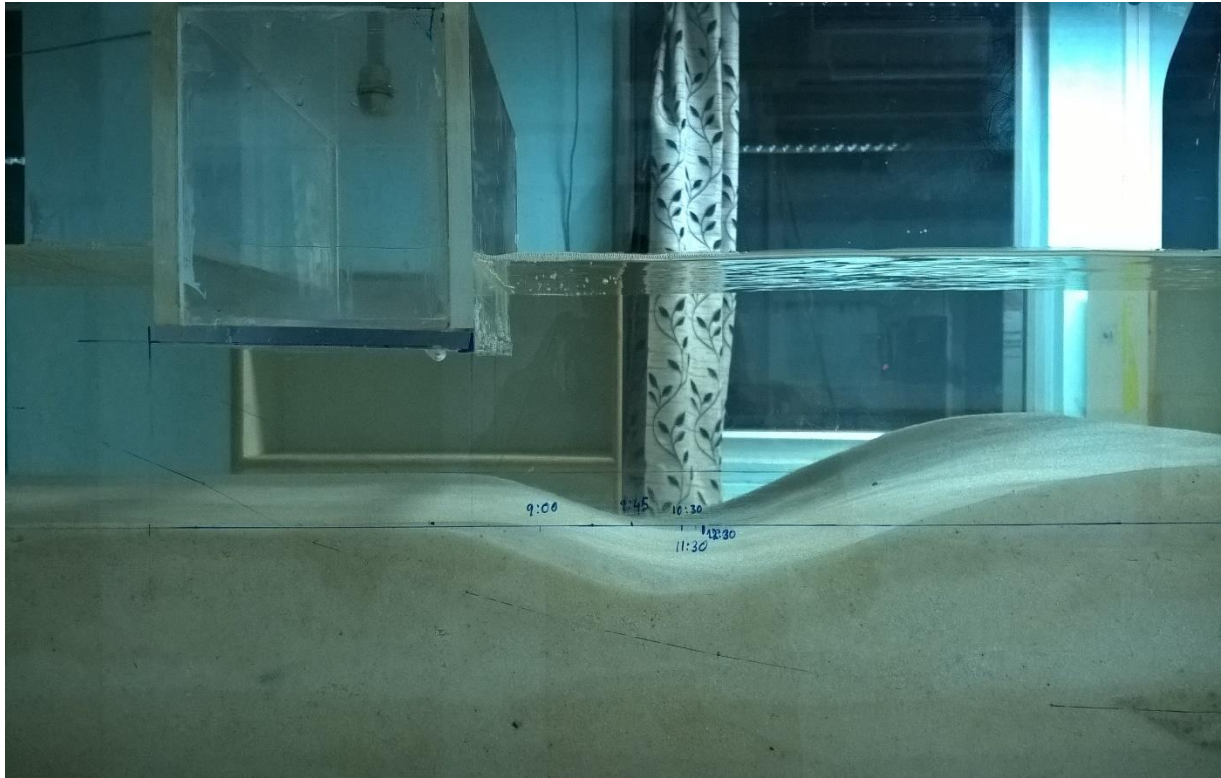
Top view

Experiment-02_(a) ,(b),(c)

Parameters	Experiment-02_(a) (keeping q constant)	Experiment-02_(b) (keeping Q constant)	Experiment-02_(c) (keeping V_a constant)
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.218,$ $\frac{H_b}{Y} = 0.6, \frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.201,$ $\frac{H_b}{Y} = 0.6, \frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.221,$ $\frac{H_b}{Y} = 0.6, \frac{V_a}{V_c} = 0.84$
Water depth	11.45 cm	14.64 cm	11.07 cm
Approach velocity	23.14 cm/s	24.1 cm/s	23 cm/s
Discharge	7.95 l/s	10.6 l/s	7.64 l/s
Bridge opening	6.87 cm	8.78 cm	6.642 cm
Max. scour location	Under the bridge	Under the bridge	Under the bridge

Experiment-03_(a) ,(b),(c)

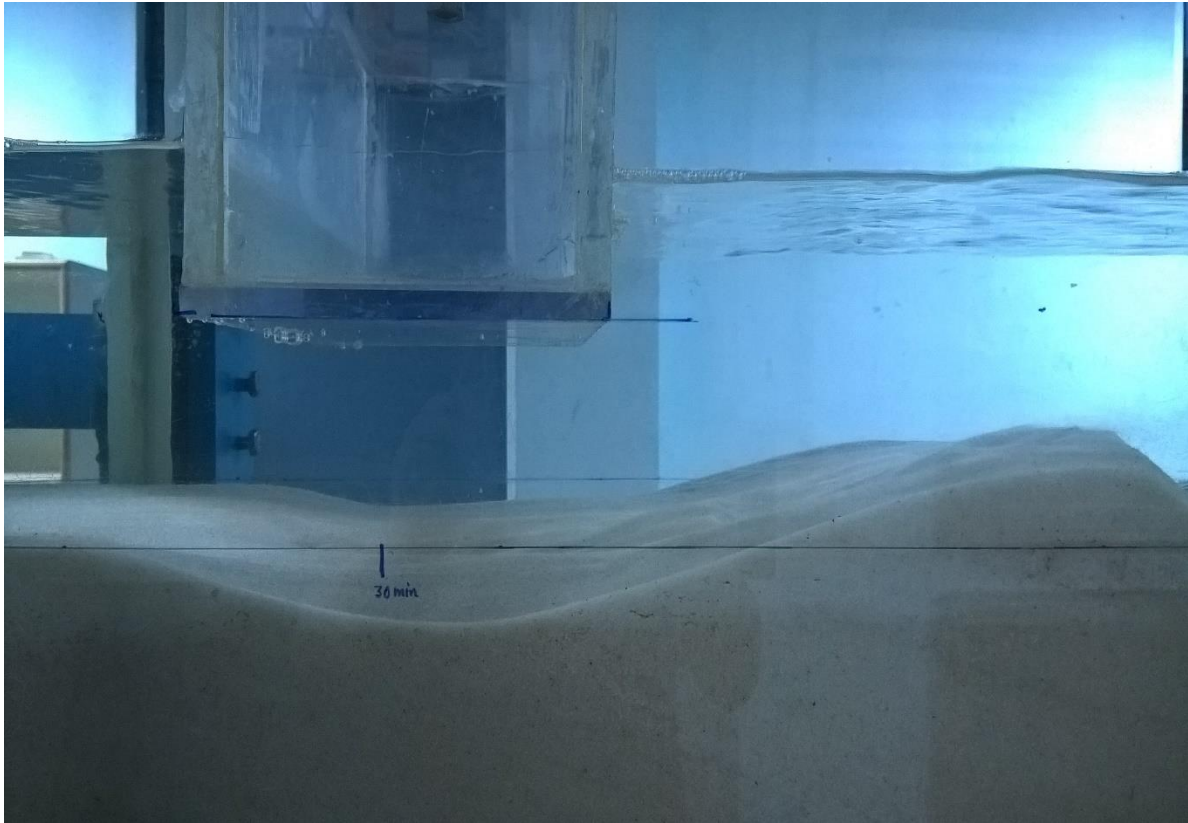
Parameters	Experiment-03_(a)	Experiment-03_(b)	Experiment-03_(c)
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.221,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.221,$ $\frac{H_b}{Y} = 0.65,$ $\frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.221,$ $\frac{H_b}{Y} = 0.7,$ $\frac{V_a}{V_c} = 0.84$
Water depth	11.07 cm	11.07 cm	11.07 cm
Approach velocity	23 cm/s	23 cm/s	23 cm/s
Discharge	7.64 l/s	7.64 l/s	7.64 l/s
Bridge opening	6.642 cm	7.2 cm	7.7 cm
Max. scour location	Under the bridge	d/s of the bridge (Near d/s edge)	d/s of the bridge



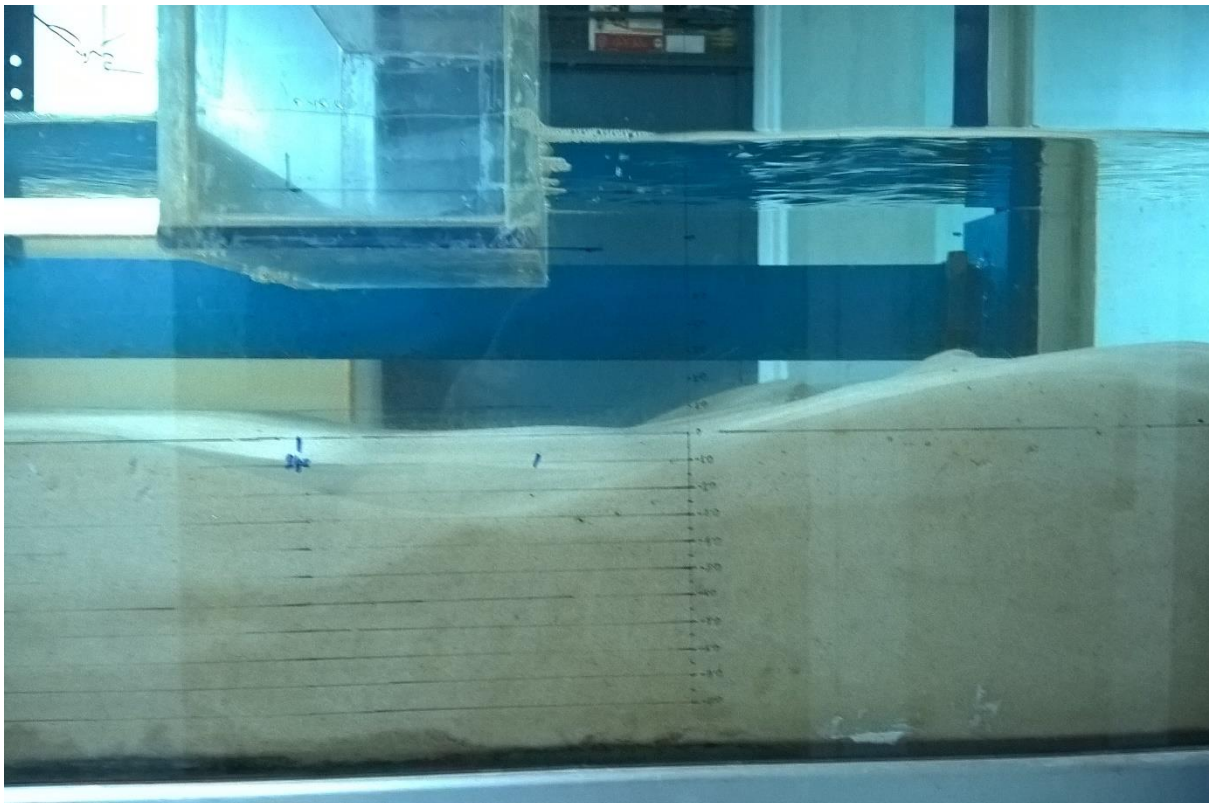
Experiment-3_(c)

Experiment-04_(a) ,(b),(c)

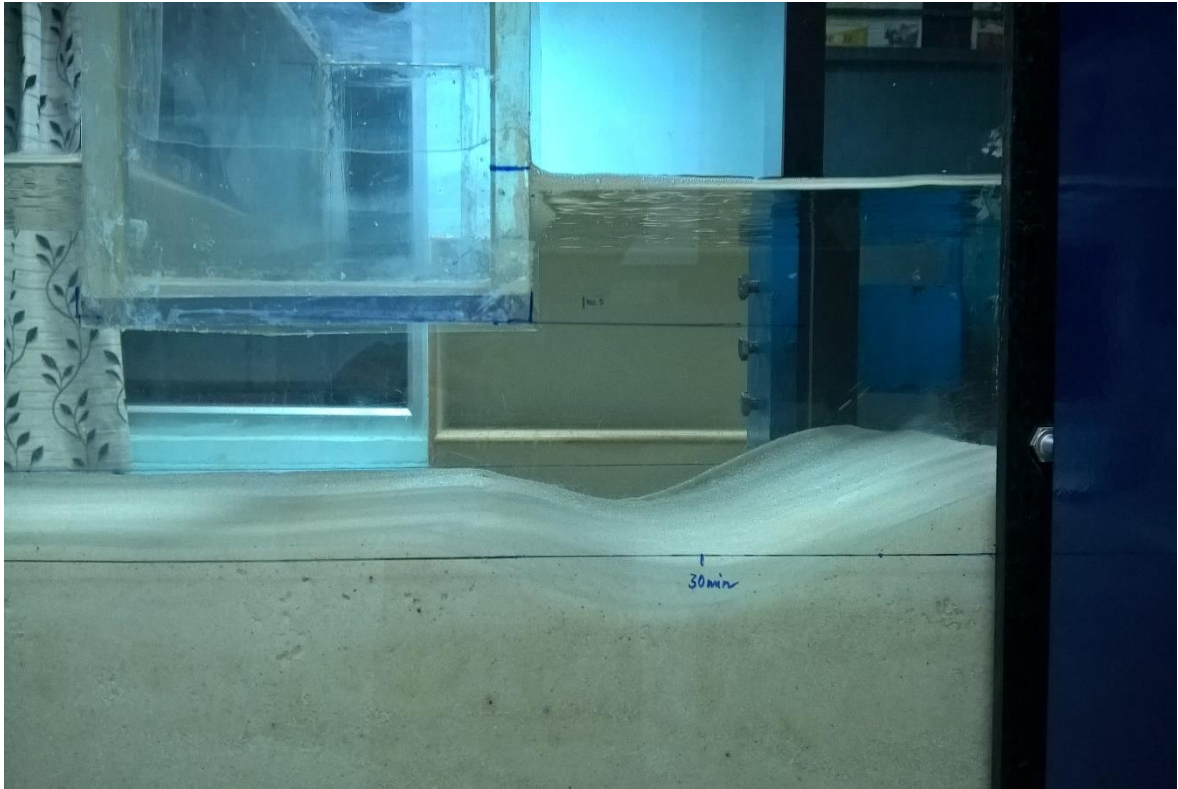
Parameters	Exp-04_(a)	Exp-04_(b)	Exp-04_(c)	Exp-04_(c)
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.216,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.82$	$\frac{V_a}{\sqrt{gY}} = 0.202,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.77$	$\frac{V_a}{\sqrt{gY}} = 0.173,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.66$	$\frac{V_a}{\sqrt{gY}} = 0.144,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.55$
Water depth (Y)	11.07 cm	11.07 cm	11.07 cm	11.07 cm
Approach velocity (V_a)	22.5 cm/s	21 cm/s	18 cm/s	15 cm/s
Discharge (Q)	7.47 l/s	6.97 l/s	5.98 l/s	4.98 l/s
Bridge opening (H_b)	6.642 cm	6.642 cm	6.642 cm	6.642 cm
Max. scour location	Under the bridge	Under the bridge (Near d/s edge)	d/s of the bridge	No scour



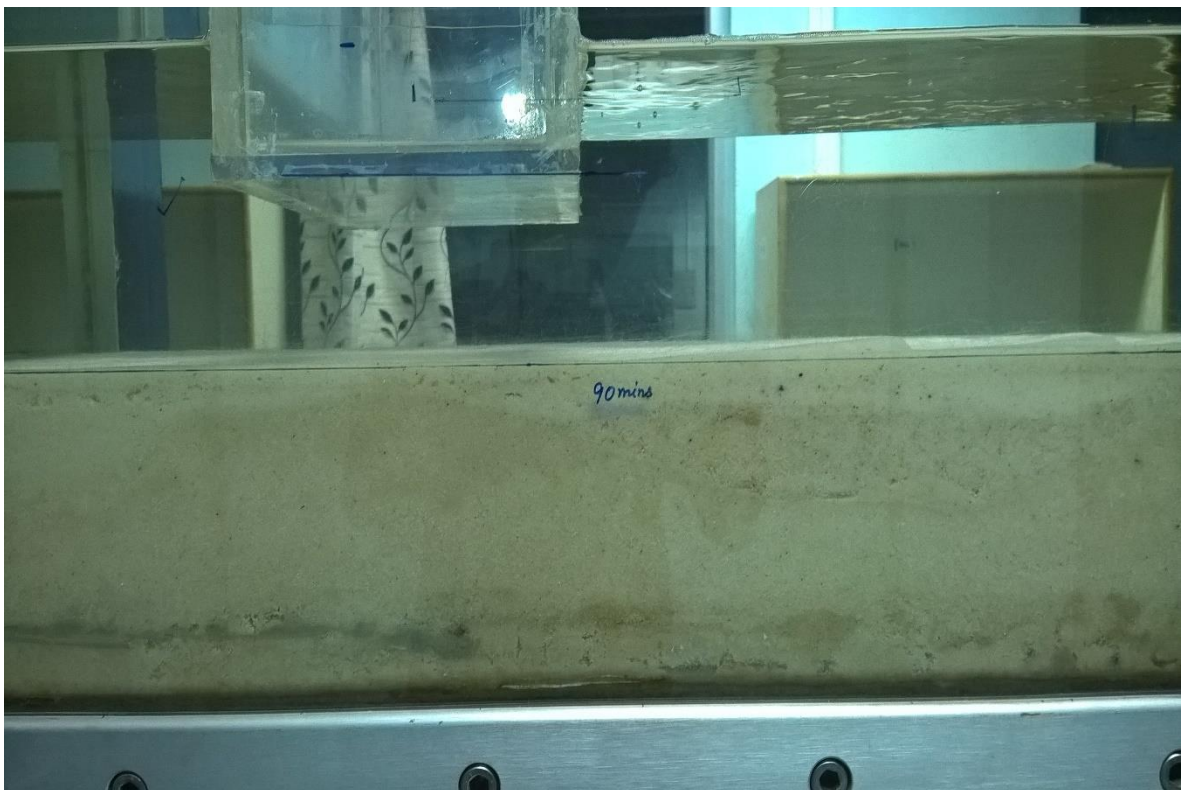
Experiment-4_(a)



Experiment-4_(b)



Experiment-4_(c)



Experiment-4_(d)

Conclusions (Exp.01-04):

1. Max scour location shifts to the d/s when bridge opening is increased keeping other parameters fixed.
2. Max scour location shifts to the d/s when approach velocity is decreased keeping water depth and bridge opening fixed.
3. It will be interesting to study whether grain size distribution of sand affects the location of maximum scour.

Experiment-05_(a) ,(b),(c)

Objective: To see the effect of water depth when discharge & bridge opening are constant.

Parameters	Experiment-05_(a)	Experiment-05_(b)	Experiment-05_(c)
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.221,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.84$	$\frac{V_a}{\sqrt{gY}} = 0.194,$ $\frac{H_b}{Y} = 0.55,$ $\frac{V_a}{V_c} = 0.76$	$\frac{V_a}{\sqrt{gY}} = 0.110,$ $\frac{H_b}{Y} = 0.38,$ $\frac{V_a}{V_c} = 0.50$
Water depth	11.07 cm	12.07 cm	17.57 cm
Approach velocity	23 cm/s	21.1 cm/s	14.5 cm/s
Discharge	7.64 l/s	7.64 l/s	7.64 l/s
Bridge opening	6.642 cm	6.642 cm	6.642 cm
Max. scour location	Under the bridge	Under the bridge	Under the bridge
Amt. of scouring	Moderate	Moderate to high	Very high
Length of scour zone	Long	Long	Very long

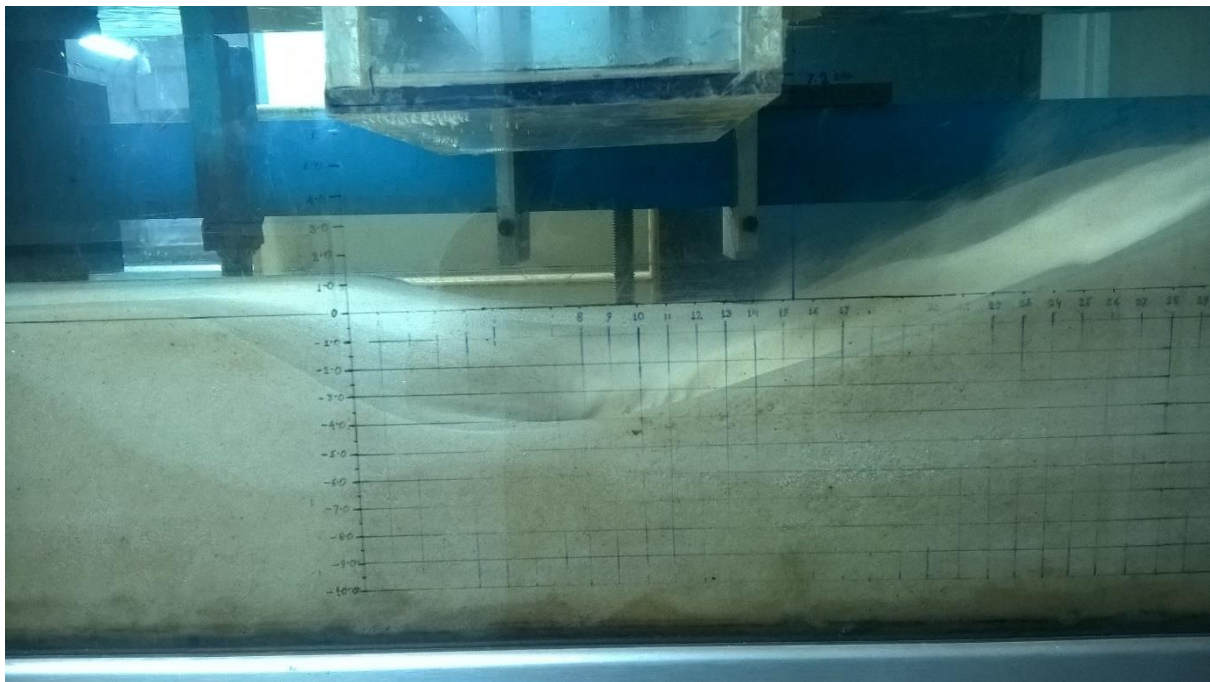
Parameters	Exp-05_(d)	Exp-05_(e)	Exp-05_(f)	Exp-05_(g)
Non-dimensional parameters	$\frac{V_a}{\sqrt{gY}} = 0.175,$ $\frac{H_b}{Y} = 0.65,$ $\frac{V_a}{V_c} = 0.78$	$\frac{V_a}{\sqrt{gY}} = 0.175,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.78$	$\frac{V_a}{\sqrt{gY}} = 0.11,$ $\frac{H_b}{Y} = 0.6 - 0.51,$ $\frac{V_a}{V_c} = 0.50$	$\frac{V_a}{\sqrt{gY}} =,$ $\frac{H_b}{Y} = 0.41,$ $\frac{V_a}{V_c} = 0.50$
Water depth (Y)	17.57 cm	17.57 cm	17.57 cm	17.57 cm
Approach velocity (V_a)	23 cm/s	23 cm/s	14.5 cm/s	14.5 cm/s
Discharge (Q)	11.43 l/s	11.43 l/s	7.64 l/s	7.64 l/s
Bridge opening (H_b)	11.4 cm	10.5 cm	10.5-8.9 cm	7.2 cm
Max. scour location	Under the bridge + d/s scour (Max scour near d/s of bridge)	Under the bridge + d/s scour (Max scour near d/s bridge edge)	d/s of the bridge	d/s of the bridge
Amt. of scouring	Moderate to low	Moderate to low	Low	Low (But high near glass)
Length of scour zone	Very long	Long	Short	Short (but long near glass)

Conclusions (Exp.05):

1. Amount of scouring mainly depends on the water depth (while bridge opening is fixed). It is more when water depth is more, because with the increment of depth weight of water gets higher, also more streamlines get distorted due to the presence of bridge.
2. On the other hand length of scour zone depends on the bridge opening. If bridge opening decreases, length of scouring increases.
3. It is difficult to comment on the effect of 'Water depth' (when discharge & bridge opening is fixed) on the location of max scour.
4. To study maximum scour location, most important parameters are bridge opening and unit discharge.



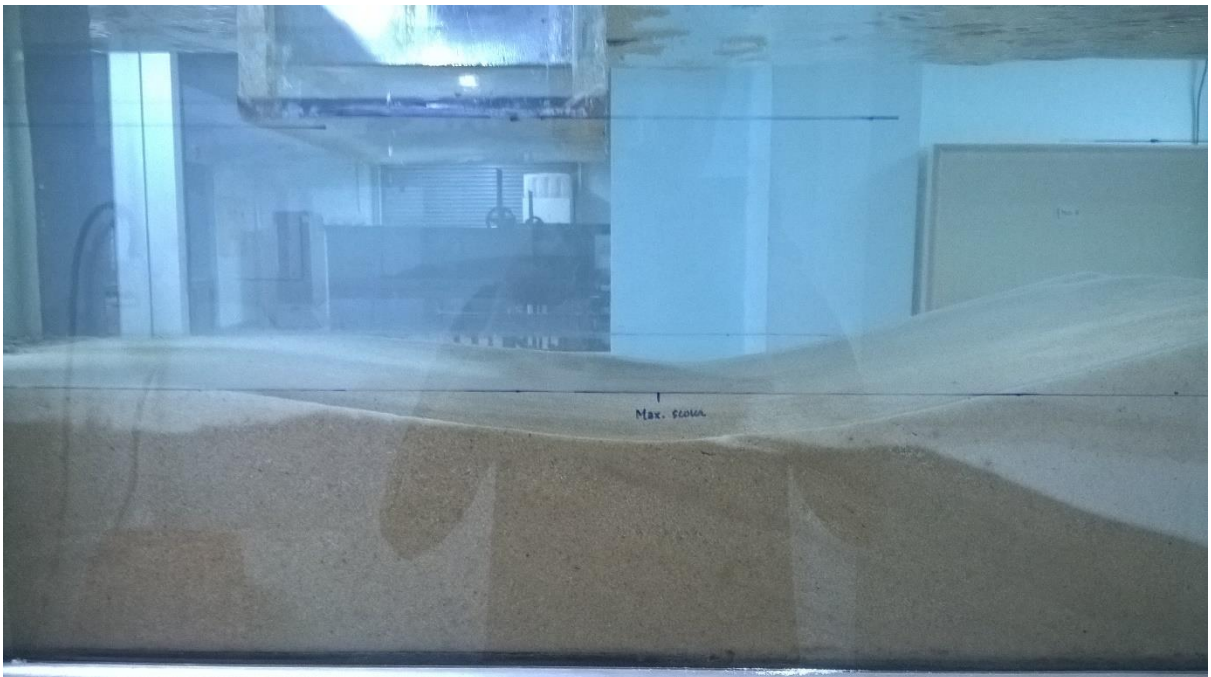
Experiment-5_(a)



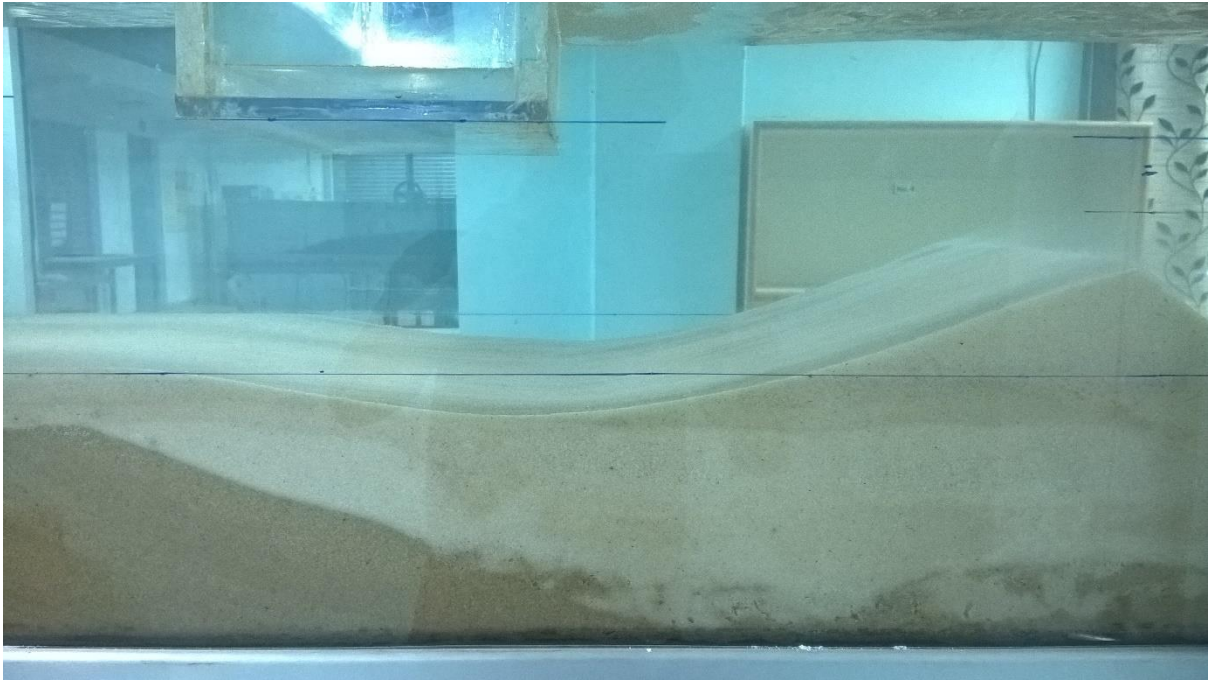
Experiment-5_(b)



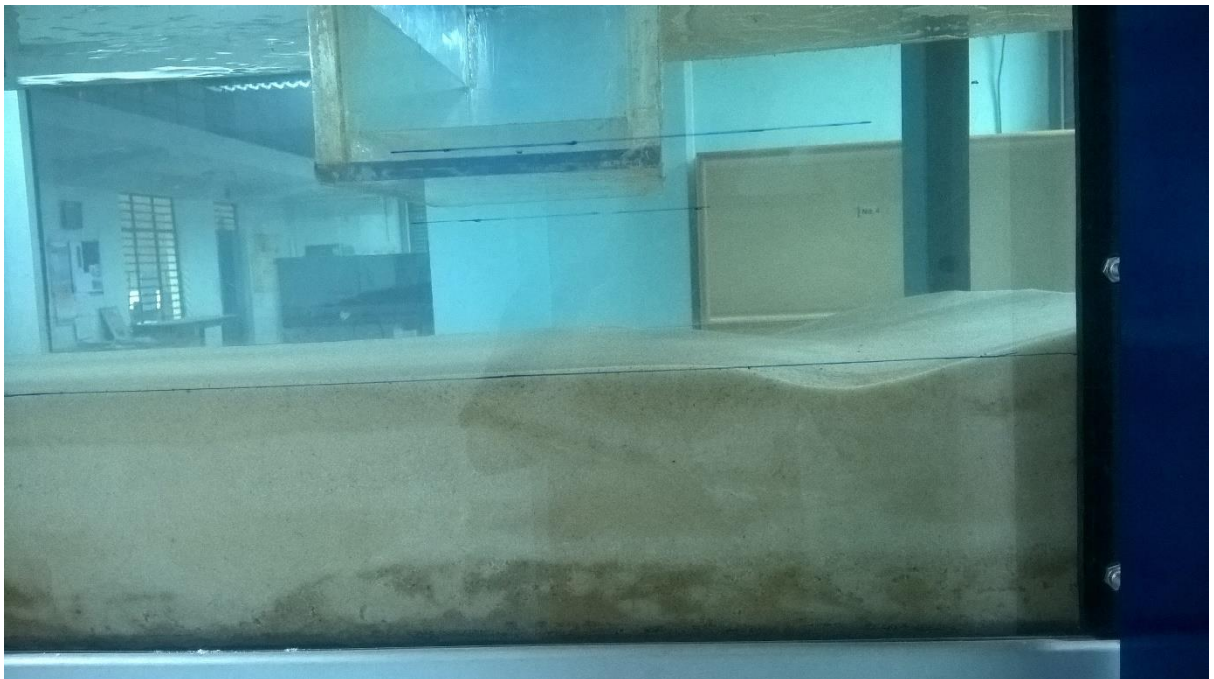
Experiment-5_(c)



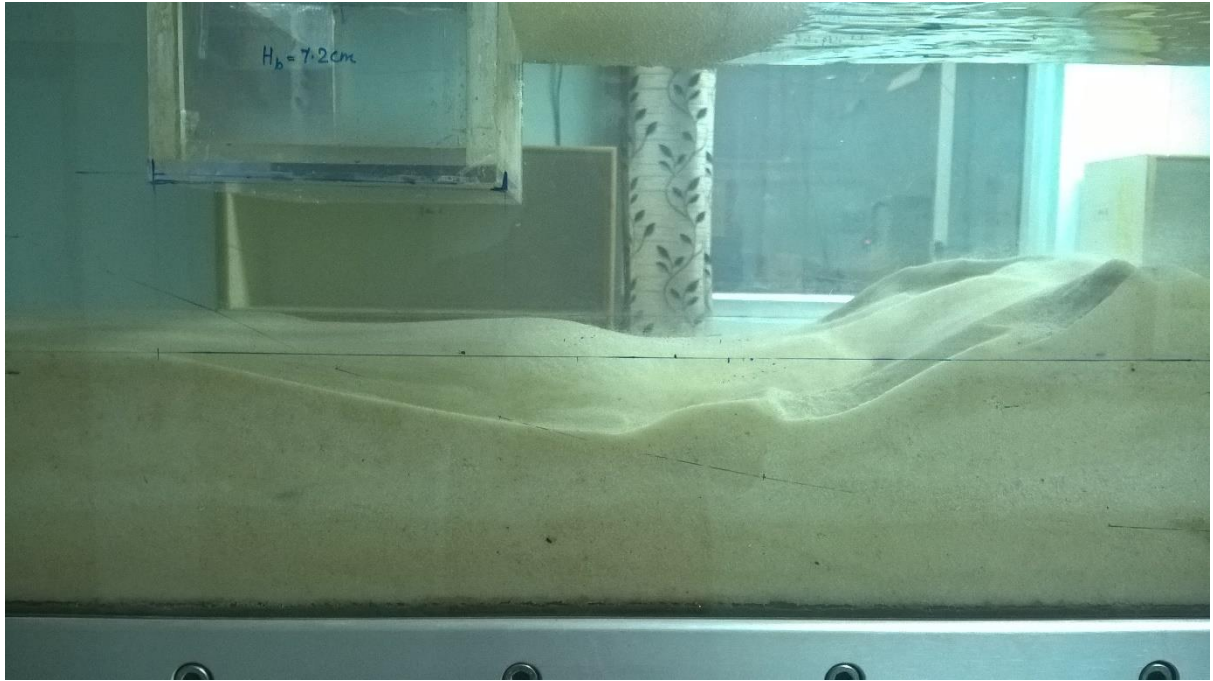
Experiment-5_(d)



Experiment-5_(e)



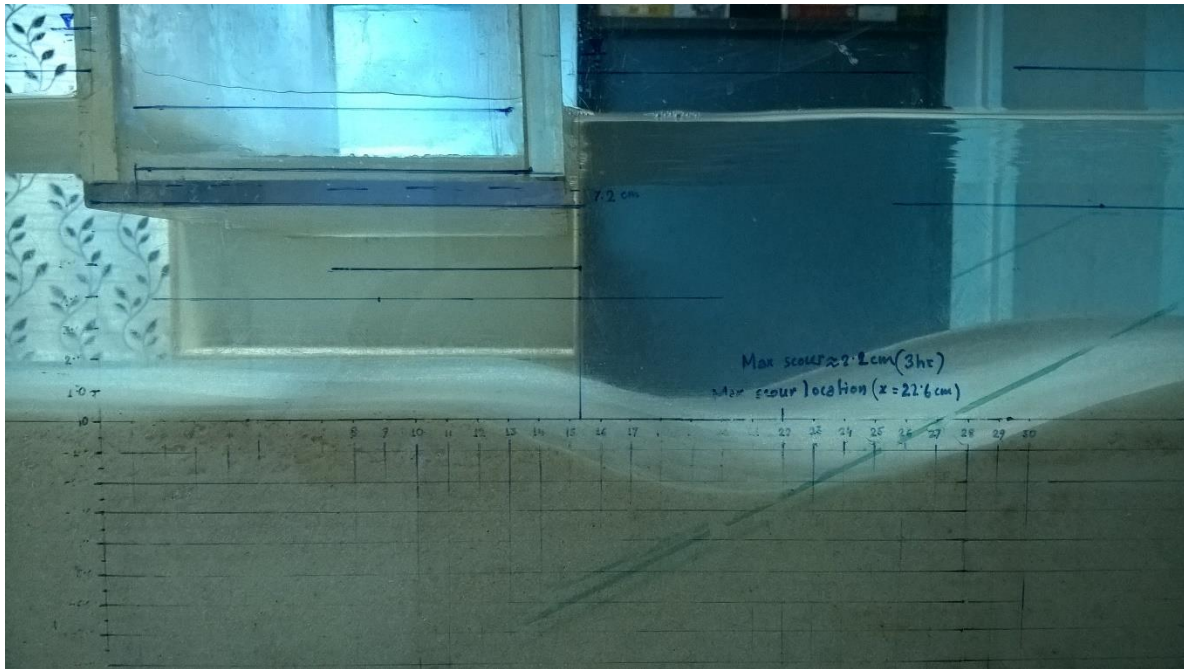
Experiment-5_(f)



Experiment-5_(g)

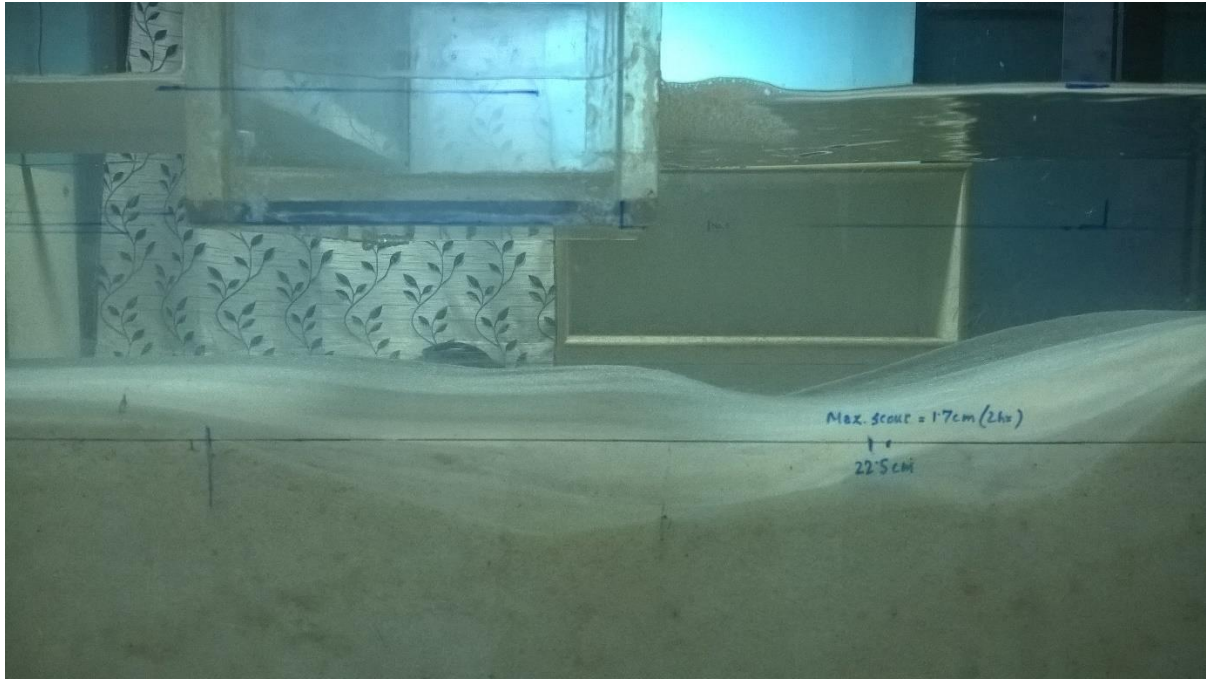
Experiment-06_(a) ,(b),(c)

Objective: To replicate the previous available results.



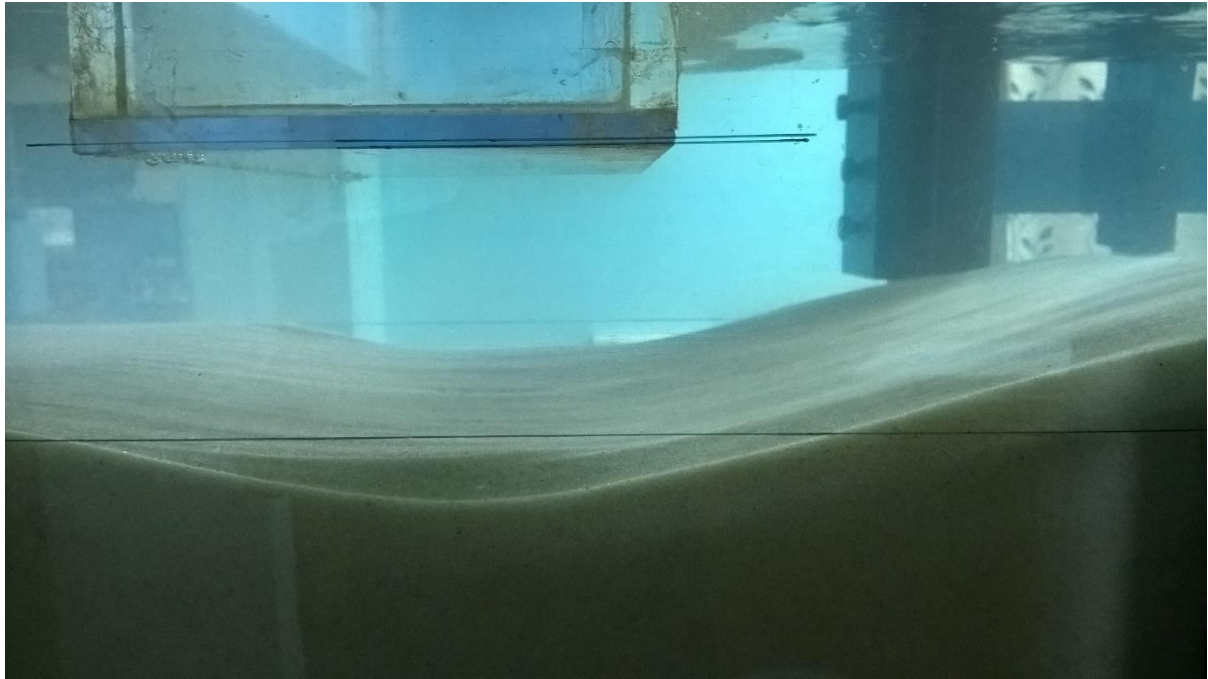
Experiment-05_SFR2

Parameters	SFR2	Experiment-05_SFR2
Non-dimensional parameters	$\frac{V_u}{\sqrt{gY}} = 0.341,$ $\frac{H_b}{Y} = 0.7,$ $\frac{V_a}{V_c} = 0.86$	$\frac{V_u}{\sqrt{gY}} = 0.341,$ $\frac{H_b}{Y} = 0.7,$ $\frac{V_a}{V_c} = 0.86$
Water depth	9.7 cm	9.4 cm
Approach velocity	23 cm/s	22.92 cm/s
Discharge	5.58 l/s	6.46 l/s
Bridge opening	6.79 cm	6.58 cm
Max. scour location (x)	21.7 cm (3hr)	22.6 cm (3hr)
Max. scouring	1.9 cm	2.2 cm



Experiment-05_SFR7

Parameters	SFR7	Experiment-05_SFR7
Non-dimensional parameters	$\frac{V_u}{\sqrt{gY}} = 0.337,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.77$	$\frac{V_u}{\sqrt{gY}} = 0.337,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.77$
Water depth	11 cm	10.7 cm
Approach velocity	21 cm/s	20.7 cm/s
Discharge	5.77 l/s	6.65 l/s
Bridge opening	6.6 cm	6.41 cm
Max. scour location (x)	24.82 cm (3hr)	22.5 cm (2hr)
Max. scouring	2.85 cm	1.7 cm



Experiment-05_LFR1

Parameters	LFR1	Experiment-05_LFR1
Non-dimensional parameters	$\frac{V_u}{\sqrt{gY}} = 0.339,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.81$	$\frac{V_u}{\sqrt{gY}} = 0.339,$ $\frac{H_b}{Y} = 0.6,$ $\frac{V_a}{V_c} = 0.81$
Water depth	12 cm	12.2 cm
Approach velocity	22.1 cm/s	21.1 cm/s
Discharge	6.63 l/s	8.14 l/s
Bridge opening	7.2 cm	7.3 cm
Max. scour location	d/s the bridge	Under the bridge
Max. scouring	-	-

Further study

1. To replicate an experiment three non-dimensional parameters $\left(\frac{V_u}{\sqrt{gY}}, \frac{H_b}{Y}, \frac{V_a}{V_c} \right)$ are considered.

If experiments are conducted in different flumes (flumes with different width), it should give the same results (same max. scour location) when these three parameters are same. These approach is quite promising as it provides same unit discharge (discharge per unit width) for different flumes. However, this is not valid when particle diameter is changed.

2. Maximum scour location may shift suddenly from the d/s to the under the bridge with the change of $\frac{V_u}{\sqrt{gY}}, \frac{H_b}{Y}, \frac{V_a}{V_c}$ for higher discharge. For lower discharge this shifting is not abrupt.

It is a continuous shifting process. Suppose if we reduce $\frac{H_b}{Y}$ keeping everything else same (for a moderate or low discharge), max scour location first shifts to the d/s edge of the bridge from a further d/s location. If the ratio is abated further, it will shift under the bridge.